Post-tuberculosis lung disease: a case definition for use in research studies

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Despite growing awareness of the substantial burden of long-term pulmonary impairment among tuberculosis survivors, marked variability in how post-tuberculosis lung disease is defined across research studies limits the comparison of findings and synthesis of evidence. To facilitate greater harmonisation within the field, we propose a case definition for post-tuberculosis lung disease for use in research studies. Conceptual aspects of this case definition were initially developed with input from a broad group of stakeholders at the 2nd International Post-Tuberculosis Symposium and were refined by the authors after the Symposium. Guiding principles for the definition include specificity, feasibility in settings with high tuberculosis disease burdens, probable relevance to long-term health outcomes, and applicability across the lifespan. The definition is designed to be used alongside, rather than instead of, study-specific definitions used to explore primary study hypotheses, and is accompanied by a reporting framework. The case definition has three components: that the individual had previous pulmonary or pleural tuberculosis disease and does not have tuberculosis disease at the time of evaluation; that the individual has, at the time of assessment, evidence of pulmonary disease with abnormalities in at least two of three clinical domains of lung function, respiratory symptoms, and chest imaging; and that the pulmonary disease manifestations should be attributable at least in part to previous tuberculosis disease. This definition is developed in the absence of data on long-term patient outcomes and will need to evolve over time in response to emerging evidence. However, we believe this proposed definition will lead to greater consistency and rigor across studies of post-tuberculosis lung disease with the goal of improving care and quality of life for millions of tuberculosis survivors worldwide.

Introduction

Approximately 10·8 million adults and children developed tuberculosis disease globally in 2023, including approximately 650 000 people living with HIV.[1] Although most individuals with drug-susceptible pulmonary tuberculosis disease are cured of their acute illness, nearly half the annual health burden from tuberculosis disease—58 million disability-adjusted life-years—are attributable to post-tuberculosis sequelae.[2] Furthermore, people previously treated for tuberculosis disease have a standardised mortality ratio approximately three times higher than the general population.[3] This elevated risk is important for the estimated 155 million tuberculosis survivors worldwide in 2020 (median age 45 years).[4]

Approximately 60% of individuals have impaired lung function after completing treatment for pulmonary tuberculosis disease.[5] Previous pulmonary tuberculosis disease is associated with chronic symptoms—including cough, dyspnoea, and reduced quality of life—and persistent imaging abnormalities that include fibrosis, bronchiectasis, and cavitation.[6–8] Advanced radiographical disease at diagnosis, repeated tuberculosis disease episodes, drug resistance, smoking, and alcohol have all been associated with post-tuberculosis lung disease, but its pathophysiology is poorly understood and there are no established interventions for treating or preventing the condition.[9,10]

At the 1st International Post-Tuberculosis Symposium in Stellenbosch, South Africa, in 2019, an international group of clinicians, researchers, and policy makers developed a consensus definition for post-tuberculosis lung disease as "Evidence of chronic respiratory abnormality, with or without symptoms, attributable at least in part to previous tuberculosis".[11] This definition encompasses the major conceptual aspects of post-tuberculosis lung disease for care, advocacy, and policy purposes. For these reasons, it deliberately prioritised sensitivity (encompassing all those with possible post-tuberculosis lung disease) over specificity (encompassing only those with true post-tuberculosis lung disease [although there is no validated diagnostic test]). The first definition was not specifically designed to be used as a formal case definition for post-tuberculosis lung disease in research studies.

Rationale

Substantial heterogeneity in how post-tuberculosis lung disease is reported between studies complicates the synthesis of findings across settings.[6,12,13] It has been defined using a variety of parameters (eg, symptoms, lung function impairment, functional testing, and imaging abnormalities), with diverse approaches and diagnostic thresholds within each domain. These varied parameters were illustrated in a 2023 meta-analysis that evaluated individuals who successfully completed pulmonary tuberculosis disease treatment and were assessed for at least one post-treatment respiratory outcome.[6] Most studies that included lung function only measured spirometry and there was variation in how pulmonary function results were reported and classified as impaired. Furthermore, criteria for defining obstruction and restriction varied, as did the reference equations to determine expected lung function values. These

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differences in the interpretation and assessment of spirometry might yield different results in the proportion of individuals defined as having post-tuberculosis lung disease and in analyses of risk factors and outcomes.[14,15] Varying instruments, protocols, analysis methods, and reporting can also be seen in studies evaluating symptoms and quality of life, exercise testing, and radiographical scores.[16]

We propose a more specific case definition of post-tuberculosis lung disease and a reporting framework for use in research studies to facilitate the comparison and synthesis of evidence across studies.

Methods

The proposed case definition and reporting framework were developed under the auspices of the 2nd International Post-Tuberculosis Symposium: an open meeting held in Stellenbosch, South Africa, in April, 2023, and attended by 152 clinicians, patients, basic scientists, epidemiologists, clinical trialists, public health specialists, and representatives from WHO, the US National Institute of Health (NIH), and individual ministries of health.[17]

Before the symposium, existing case definitions and clinical staging systems from the fields of respiratory medicine and infectious diseases were reviewed to identify potential frameworks that could be used to classify post-tuberculosis lung disease. These frameworks included definitions for tuberculosis-associated immune reconstitution inflammatory syndrome,[18] tuberculosis meningitis,[19] Duke criteria for infective endocarditis,[20] idiopathic pulmonary fibrosis,[21,22] long COVID,[23] asthma, and chronic obstructive pulmonary disease (COPD).[24,25]

On the first day of the conference, multiple-choice questions were posed to all attendees at the meeting's introductory session, with live voting used to gauge interest in the need for a more specific case definition of post-tuberculosis lung disease. Small-group workshops during the Symposium further explored interests and priorities. Post-conference live voting on the final day of the Symposium identified areas of consensus (table).

Following the Symposium, the authors of this Personal View, which include a tuberculosis survivor and specialists in respiratory and infectious disease medicine and paediatrics, continued to discuss and refine the case definition for post-tuberculosis lung disease. In addition, we presented this definition at a plenary session at the 3rd International Post-Tuberculosis Symposium in Stellenbosch, South Africa, in April, 2025, and feedback from that session was considered for inclusion.

Guiding principles for a case definition

Several concepts and principles emerged during the 2nd Symposium workshops and guided the formulation of this case definition.

Specificity: a research definition for post-tuberculosis lung disease should prioritise specificity to focus on individuals with respiratory disease that is causally attributable to previous tuberculosis disease. Although there is no current consensus standard for diagnosis of post-tuberculosis lung disease, conceptual threats to specificity include the absence of previous tuberculosis disease (eg, misdiagnosis), ongoing tuberculosis disease (eg, microbiological failure or recurrent tuberculosis disease), limitations that stem from non-pulmonary processes (eg, respiratory muscle weakness or deconditioning), or pathology related to non-tuberculosis respiratory disease (eg, smoking-related COPD).

Feasibility in settings with a high tuberculosis disease burden: given the high burden of tuberculosis disease in low-income and middle-income countries (LMICs), it was agreed that the definition should not require advanced diagnostics or costly equipment. A research definition that is not readily implementable in routine clinical care would hinder translation of research findings into practice and decrease real-world effects on patient outcomes.

Probable relevance to long-term health outcomes: although few studies have established how respiratory measures relate to long-term outcomes among tuberculosis survivors, the case definition should focus on respiratory pathology that could plausibly affect future health. Fulfilment of this principle largely requires reliance on data from other chronic respiratory diseases.

Applicability across the lifespan: estimates indicate that up to 12% of incident tuberculosis disease occurs among children and adolescents, many of whom have respiratory sequelae.[26,27] The case definition should therefore support the generation of an evidence base for both children and adults.

Complementary to primary study hypotheses: although some studies might use this post-tuberculosis lung disease case definition as a primary outcome, it is not intended to replace hypothesis-driven endpoints and can be reported as a secondary endpoint in supplemental materials.

Imperative to proceed in the absence of sufficient evidence: diagnostic criteria for many diseases, including respiratory conditions such as COPD and asthma, evolve over time.[28,29] We recognise that existing post-tuberculosis lung disease measures have not been adequately validated with long-term outcomes in tuberculosis survivors, but hypothesise that a shared case definition will accelerate data harmonisation, thus supporting future validation of respiratory measures, methods, and cut points across diverse settings and cohorts. We anticipate that this definition will be refined over time.

We propose that authors complete a reporting framework alongside the case definition when studying post-tuberculosis lung disease to ensure transparency about the approaches used and to support further evidence synthesis across studies.

Proposed case definition and key considerations

There are three major components to this post-tuberculosis lung disease case definition: antecedent conditions enabling a focus on patients who are truly post-tuberculosis

	Response (%)
Pre-conference questions	
Do you think there is a need to agree on a more specific definition (or definitions) of post-tuberculosis lung disease for use within research studies?	
Yes	66/88 (75%)
Maybe	14/88 (16%)
No	8/88 (9%)
What would your priorities be in a research definition of post-tuberculosis lung disease? (Check all that apply)	
Being sensitive, capturing as many with actual post-tuberculosis lung disease as possible, while accepting the inclusion of some without actual post-tuberculosis lung disease (false positives)	35/88 (40%)
Being specific, capturing only those with clear evidence of lung damage or respiratory impairment attributable to tuberculosis, while accepting that some with actual post-tuberculosis lung disease will be missed (false negatives)	44/88 (50%)
Having a severity grade, such as mild, moderate, and severe	66/88 (75%)
Having a certainty grade, such as possible, probable, and confirmed	51/88 (58%)
Being adaptable for use in resource-limited settings	58/88 (66%)
Other	6/88 (7%)
Which tools should be used to define post-tuberculosis lung disease within the research definition? (Check all that apply)	
Respiratory symptoms (eg, cough, shortness of breath, chest pain, or wheeze)	77/88 (88%)
Systemic symptoms (eg, fatigue or low energy)	46/88 (52%)
Respiratory quality of life score (eg, St George's Respiratory Questionnaire)	58/88 (66%)
Overall quality of life score (eg, EQ-5D-3L)	41/88 (47%)
Exercise testing (eg, sit to stand, 6-min walk, or incremental shuttle)	68/88 (77%)
Pulse oximetry	40/88 (45%)
Spirometry	66/88 (75%)
Broader lung function testing (eg, lung volumes or gas transfer)	41/88 (47%)
Chest x-ray	57/88 (65%)
CT	40/88 (45%)
Echocardiography	19/88 (22%)
Other not listed above – please list in comments	10/88 (11%)
(Table continues in next column)	
(Continued from previous column)	
Which of the following do you think is the single most important measure for determining if a patient has clinical evidence of post-tuberculosis lung disease?	
All equally important	29/88 (33%)
Respiratory symptoms	21/88 (24%)
Lung function	21/88 (24%)
Radiology	8/88 (9%)
Other	5/88 (6%)
Unsure	4/88 (5%)
Post-conference questions	
Do you think there is a need for a standardised reporting framework for post-tuberculosis lung disease to use within research studies? This framework is not meant to replace hypothesis-drive outcomes but would provide a way to harmonise post-tuberculosis lung disease reporting	
Yes	79/79 (100%)
No	0
Do you think that this framework should include a severity grade?	
Yes	70/76 (92%)
No	6/76 (8%)
Do you think that this framework should include a certainty grade?	
Yes	72/82 (88%)
No	10/82 (12%)

EQ-5D-3L=three-level, five-dimensional, EuroQoL questionnaire. QoL=quality of life.

Table: **Input on the definition of post-tuberculosis lung disease at the 2nd International Post-Tuberculosis Symposium**

(they previously had tuberculosis disease), assessments indicating current pulmonary disease and its severity, and efforts to assess alternative causes of lung disease (panel 1).

(1) Antecedent requirements

The individual previously had pulmonary or pleural tuberculosis and does not currently have active tuberculosis disease

Antecedent requirements are intended to increase the likelihood that the individual had previous pulmonary or pleural tuberculosis disease and does not have tuberculosis disease requiring further antitubercular therapy at the time of assessment. Without these conditions, some individuals programmatically defined as having successfully completed tuberculosis disease treatment might actually not be post-tuberculosis, either because they never had tuberculosis disease or because they require additional antitubercular therapy before presenting symptoms and signs can truly be considered sequelae, which are defined as an after-effect of a disease, condition, or injury. Although the optimal timing of post-tuberculosis lung disease assessments is unknown, we suggest that evaluation to exclude tuberculosis disease be done as close as possible to the evaluation of post-tuberculosis lung disease.

Inclusion of individuals with either bacteriologically confirmed or clinically diagnosed previous tuberculosis disease

Inclusion of individuals with both bacteriologically confirmed and clinically diagnosed tuberculosis disease aligns the definition with existing WHO and NIH diagnostic criteria.[30,31] We recognise that some individuals who have clinically diagnosed tuberculosis disease will not have tuberculosis disease. Although requiring bacteriological confirmation would likely increase the certainty of previous tuberculosis disease, it would risk excluding a large proportion of individuals who are diagnosed with clinically diagnosed tuberculosis disease, many of whom live in LMICs, are children, or are individuals with a low bacterial burden.[1] Of note, as

Panel 1: **Proposed case definition for post-tuberculosis lung disease for use in research**

There are three components to this case definition that must be satisfied for a diagnosis of post-tuberculosis lung disease.

(1) Antecedent requirements

Both of the following should be met using medical record documentation (ie, not self-report):

- The individual previously had pulmonary or pleural tuberculosis, which was successfully treated.
  - For adults, previous tuberculosis must fulfil WHO criteria for bacteriologically confirmed or clinically diagnosed pulmonary or pleural tuberculosis that led to the decision to treat with a course of antitubercular therapy.[30] Pulmonary tuberculosis disease classification is based on the WHO definition of any bacteriologically confirmed or clinically diagnosed tuberculosis disease involving the lung parenchyma or tracheobronchial tree.[30]
  - If the previous pulmonary or pleural tuberculosis diagnosis was not bacteriologically confirmed, in addition to the clinical tuberculosis diagnosis, there must be radiographical evidence of pulmonary or pleural involvement. Evidence can come from imaging studies either at diagnosis or from a post-tuberculosis treatment evaluation. Evidence includes but is not limited to bronchiectasis, cavitation, nodules, fibrosis, lobar collapse, ground-glass opacities, consolidation, pleural effusion, pleural based lesions, and pleural thickening.
  - Children must fulfil diagnostic criteria for confirmed or unconfirmed intrathoracic tuberculosis using the US National Institutes of Health clinical tuberculosis case definitions.[31]
- The individual should not have active tuberculosis disease, meaning tuberculosis requiring additional antitubercular therapy at the time of post-tuberculosis lung disease assessment. Evidence for the absence of current active tuberculosis disease includes:
  - Absence of a current need for additional antitubercular therapy as established by a clinician or clinician investigator, and either negative microbiological assessments sufficient to establish a WHO-defined cure[32] or WHO-defined tuberculosis treatment completion.[32]

(2) Current requirements

To meet this research definition for post-tuberculosis lung disease, individuals must have abnormalities in at least two of the following domains at the time of post-tuberculosis lung disease assessment: lung function, symptoms, and imaging.

If an individual has abnormalities documented within only one of the three domains, they will be considered to have a post-tuberculosis lung abnormality.

Impaired lung function

- Defined as having a value less than or equal to the 5th percentile of a reference population (z-score ≤–1·645) for at least one of forced expiratory volume in 1 s (FEV_1), forced vital capacity (FVC), FEV_1 to FVC ratio, haemoglobin-adjusted diffusion capacity, or total lung capacity.
- Global Lung Function Initiative-appropriate reference equations for the lung function parameters listed above should be used to define the expected range of values in healthy individuals.[29] The 5th and 95th percentile limits (–1·645 and +1–645 z-score) of the healthy population should be used to establish impairment.[29]

Chronic respiratory symptoms

- Presence of at least one of the following for at least 8 weeks for adults or 4 weeks for children:
  - Cough with or without sputum production or haemoptysis.
  - Dyspnoea or breathlessness.
  - Wheezing.
  - Limitation in activities attributed to respiratory impairment.
- Symptoms should have either begun or worsened during or after the period of active tuberculosis disease.

Imaging criteria

- Presence of radiographical abnormalities observed after tuberculosis treatment completion that plausibly explain, at least in part, the patient's respiratory symptoms or impairment and are consistent with previous pulmonary or pleural tuberculosis, including but not limited to:
  - Fibrosis.
  - Cavitation or parenchymal destruction.
  - Pleural thickening.
  - Bronchiectasis.
  - Volume loss.

(3) The radiographical, functional, and symptomatic pulmonary disease manifestations should be attributable, at least in part, to previous tuberculosis disease

Alternative, non-tuberculosis pulmonary conditions should not completely explain the person's pulmonary disease manifestations or risk factors (eg, smoking).

detailed below, this definition includes asymptomatic tuberculosis disease, including early subclinical disease.

Requirement of radiological evidence of previous pulmonary or pleural tuberculosis disease if clinically diagnosed

Given the potential for inaccurate clinical diagnoses, it was felt that individuals with previous clinically diagnosed tuberculosis disease should also have radiographical evidence of pleural or pulmonary involvement consistent with tuberculosis disease. This evidence could be drawn from chest x-ray or CT imaging completed either during or after tuberculosis disease to provide evidence that the individual's previous tuberculosis disease involved the lung parenchyma, airways, or pleura.

The decision to include pleural tuberculosis disease but not other forms of extrapulmonary tuberculosis

disease that might affect respiratory function is due to the close relationship of the pleura and the lungs, anatomically and functionally, and on the definition's explicit focus on lung disease. Spinal, pericardial, lymph node, laryngeal, or other extrapulmonary forms of tuberculosis disease are not considered antecedent forms of tuberculosis disease in this definition.

Inclusion of individuals with a history of treatment success with either microbiological cure or clinical treatment completion

Most individuals living in settings with high tuberculosis disease burdens who complete standard treatment do not meet criteria for tuberculosis cure, primarily because cultures on or after treatment are not done.[32–34] Requiring all individuals to have microbiological evidence of cure would therefore exclude most individuals and studies from LMICs. The inclusion of individuals with recorded treatment success, defined either as microbiological cure or treatment completion, aligns with the decision to include both bacteriologically confirmed and clinically diagnosed tuberculosis disease. Confirming a microbiological cure could be particularly important in some situations, such as drug-resistant disease, in which risk of microbiological treatment failure and post-tuberculosis lung disease are both high.[34] Individuals who do not complete or never receive tuberculosis disease treatment might still be classified as post-tuberculosis—for example via self-cure—but would not meet criteria for this definition.

Use of either microbiological cure or clinical assessment to evaluate active tuberculosis disease at the time of post-tuberculosis lung disease assessment

Because study enrolment can occur years after tuberculosis disease treatment completion, individuals should be assessed for active tuberculosis disease at the time of evaluation for post-tuberculosis lung disease using microbiological or clinical assessments, or, ideally, both. This assessment is particularly important because symptoms and signs of tuberculosis disease, post-tuberculosis lung disease, and other respiratory infections often overlap. Although negative sputum cultures increase the likelihood that an individual being assessed for post-tuberculosis lung disease does not have tuberculosis disease, cultures are often not available in high-burden settings and, even if available, should be paired with clinical assessment. Importantly, positive results on DNA-based tests among individuals with previous tuberculosis disease are not specific for ongoing active tuberculosis disease and should be interpreted with caution.[35,36]

(2) Current requirements

At the time of post-tuberculosis lung disease assessment, there must be evidence of pulmonary or pleural disease, with abnormalities in at least two of the following clinical domains: lung function, respiratory symptoms, and imaging. These requirements are intended to focus on respiratory sequelae plausibly related to previous tuberculosis disease that are likely to adversely affect health and wellbeing (figure). Although assessments for post-tuberculosis lung disease range from simple to complex,[16] attendees of the 2nd Symposium prioritised the evaluation of respiratory symptoms for inclusion in a case definition for post-tuberculosis lung disease, in part due to their relative ease of measurement and patient-centric nature. Evaluation of lung function, exercise testing, and imaging were also considered important, but there was no clear consensus on a single priority among them. Therefore, multiple complimentary evaluations were included (table).

General approach

Evidence to date suggests that lung function continues to change in the year after tuberculosis disease treatment completion.[37] It is therefore possible that assessments done early after treatment completion might overestimate the burden of fixed residual damage. Measurements made years after tuberculosis disease, however, could be influenced by non-tuberculosis exposures and illnesses. In addition, different studies can be interested in diseases

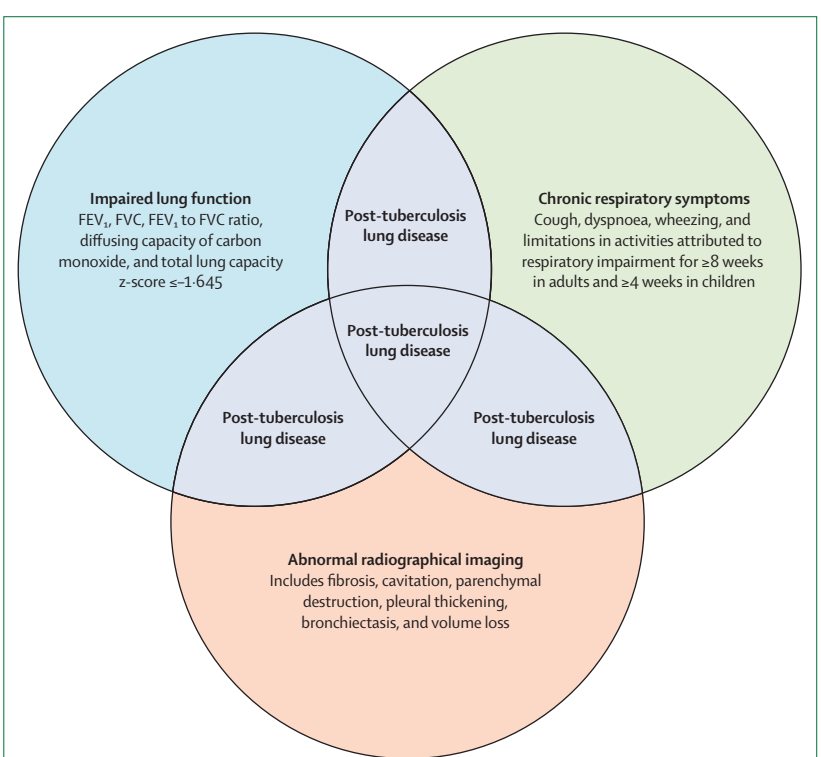


Figure: **Post-tuberculosis lung disease research case definition**
Sizes of the categories are conceptual and not configured according to prevalence or burden. Not all items listed in each domain are needed to meet criteria for that domain. Only two domains are required for a post-tuberculosis lung disease definition. FEV_1=forced expiratory volume in 1 s. FVC=forced vital capacity.

and interventions at different timepoints after treatment completion. Furthermore, post-tuberculosis lung disease assessments at the time of tuberculosis disease treatment completion for research studies should be practical and scalable in large treatment programmes. Thus, rather than specify a particular period within which post-tuberculosis lung disease must be defined, it was agreed that investigators should report the timing of their evaluations relative to the completion of tuberculosis disease treatment in the reporting framework (panel 2), as previously done.[37]

In published studies and meta-analyses of respiratory sequelae of tuberculosis disease, the most common approaches have involved evaluation of lung function via spirometry, assessment of respiratory symptoms, and imaging.[5,6,9,13,26] These three types of disease assessments are complementary measures that capture different aspects of respiratory health using methods already commonly used by those working in the field.

Requiring abnormalities in more than one clinical domain is consistent with diagnostic approaches to respiratory diseases, including asthma, interstitial lung diseases, and COPD.[22,24,25] The relationship between abnormalities in these domains and long-term patient outcomes needs validation in post-tuberculosis lung disease. For example, it is unclear whether isolated radiographical abnormalities or lung function deficits reflect subclinical pathology that will progress, remain static, or resolve over time. However, we anticipate that these data, once available, can be used to refine this section of the definition.

The term post-tuberculosis lung abnormality is proposed to account for abnormalities restricted to a single domain, whether in lung function, symptoms, or imaging, with the eventual goal of establishing whether these changes are associated with worse clinical outcomes over time.

Lung function criteria

Spirometry has a clear role in the assessment of post-tuberculosis lung disease in research. It is safe and feasible, particularly later during tuberculosis disease treatment and, although it requires staff training and equipment, spirometry has been successfully conducted across LMICs.[38] The key spirometric outputs of forced expiratory volume in 1 s (FEV_1), forced vital capacity (FVC), and FEV_1 to FVC ratio all have associations with survival and other clinical outcomes in non-tuberculosis respiratory diseases.[39–42] The multiple additional measures of lung function (eg, forced expiratory flow between 25% and 75% of the FVC and forced expiratory flow at 50% of the FVC) generated by spirometry are not widely assessed or reported, nor have they been consistently incorporated into the diagnosis of other chronic diseases, such as COPD. We thus opted to limit the qualifying spirometric parameters in this definition to FEV_1, FVC, and FEV_1 to FVC ratio.

Measurements of the diffusing capacity of carbon monoxide and lung volumes are not widely available in settings with high tuberculosis disease burdens and are less widely used in studies of post-tuberculosis lung disease. However, the diffusing capacity of the lung measures the transfer of gas across the alveolar wall and is affected by multiple factors that could plausibly be affected in post-tuberculosis lung disease, including the alveolar surface area, capillary blood flow, haemoglobin concentration, and alveolar membrane thickness.[43] Several studies suggest that individuals treated for tuberculosis disease could have reduced diffusing capacity.[44–46] Lung volumes can be measured with body plethysmography, multiple breath washout, and inert gas dilution;[47] can identify the residual volume, quantify air trapping, and other processes not evident on spirometry alone; and are required to definitively confirm restrictive lung disease.[48,49] Both lung volumes and diffusing capacity were included as parameters that could satisfy the proposed case definition, given the complementary information they provide to spirometry.

Of note, the post-tuberculosis lung disease case definition is based on impairment in individual parameters and does not require the designation of restrictive, obstructive, or mixed patterns of lung function impairment. The proportion of participants falling into each of these categories should, however, be described in the reporting framework (panel 2) using European Respiratory Society (ERS)–American Thoracic Society (ATS) interpretation guidelines.[29] Finally, as lung function assessments can be challenging in children and adolescents, further investigations into alternative approaches—such as oscillometry—in this population are needed.[27] Oscillometry measures lung function using sound waves at different frequencies to quantify resistance and reactance of the lungs during tidal breathing. It has been used for other respiratory diseases, such as COPD and asthma, particularly in young children for whom spirometry is not possible.[50] Given the few data available for oscillometry in post-tuberculosis lung disease, we opted not to include it in the current case definition.

Lung function data can be presented in multiple ways (eg, percent predicted values, lower limit of normal, and Z-scores) and can use different reference standards, all of which can influence classification and severity of impairment and add complexity to comparisons across studies.[51] In accordance with the 2021 ERS–ATS guidelines, we propose that all studies of post-tuberculosis lung disease report Global Lung Initiative race-neutral standards and Z-scores.[29] Z-score cutoffs can also be used to indicate the severity of impairments in lung function, as described later. Using these standards should facilitate the comparison of lung function results across studies.[51] Similarly, we recommend, at minimum, the use of prebronchodilator values for all measures while allowing for the documentation of bronchodilator responsiveness

Panel 2: **Reporting framework for research studies on post-tuberculosis lung disease**

Report n (%) with non-tuberculosis respiratory diseases considered as possible alternative explanations for respiratory sequelae. For current requirements, repeat relevant reporting components for all major timepoints assessed in the study. For the full reporting framework, see the appendix (pp 1–4).

Antecedent requirements

- Report most recent previous tuberculosis disease classification according to WHO definitions for adults and US National Institutes of Health definitions for children
- Report the site of most recent previous tuberculosis disease (eg, pleural or pulmonary)
- Report drug susceptibility of the most recent previous pulmonary or pleural tuberculosis disease episode, including whether this is unknown
- Report treatment outcome of most recent previous pulmonary or pleural tuberculosis disease episode using WHO definitions
- Report the time interval between tuberculosis disease treatment completion and post-tuberculosis lung disease assessment in the study
- Report the types of tuberculosis disease assessments done at the time post-tuberculosis lung disease was assessed in the study

Current requirements: evidence of pulmonary or pleural disease at the time of post-tuberculosis lung disease assessment

- Report how many individuals had lung function tests done
- Report how many individuals had one or more of the symptoms listed in the research definition assessed
- Report how many individuals had post-tuberculosis chest imaging done
- Report the main lung function parameters assessed
- Report lung function results using global lung function initiative reference equations and standard American Thoracic Society–European Respiratory Society interpretive criteria, including continuous z-scores, categorisation (eg, restrictive, obstructive, or normal), and severity
- Report whether individuals had resting, chronic resting, or exertional hypoxaemia
- Report how many individuals had any of the following symptoms for at least 8 weeks at the time of post-tuberculosis lung disease assessment: cough, wheezing, dyspnoea, or limitation of activities
- Report results of the modified Medical Research Council dyspnoea score for symptom severity
- Report summary measures for other questionnaire (ie, quality of life, symptoms, and others) scores
- Report any results of imaging evaluations done as part of post-tuberculosis lung disease assessments: report how it was established if imaging abnormalities were present or absent as implemented in the categorisation of patients for post-tuberculosis lung disease determination
- Report number of individuals in each of the following post-tuberculosis lung disease or post-tuberculosis lung abnormality mutually exclusive categories
  - Post-tuberculosis lung disease
    - Lung function and symptoms and imaging
    - Lung function and symptoms without imaging
    - Lung function and imaging without symptoms
    - Imaging and symptoms without lung function
  - Post-tuberculosis lung abnormality
    - Only lung function
    - Only symptoms
    - Only imaging
  - Other
    - No post-tuberculosis lung disease or post-tuberculosis lung abnormality

in the reporting framework. This recommendation is consistent with clinical guidelines for spirometry, which stress the uncertain relationships between responsiveness and outcomes in respiratory disease and the absence of consensus on assessment protocols.[29,52]

Symptom criteria

Respiratory symptoms are patient-centric, inexpensive to collect, and are frequently used in clinical and research descriptions of post-tuberculosis lung disease.[13] However, there are various limitations in their use to define post-tuberculosis lung disease: adverse respiratory outcomes have been observed in asymptomatic tuberculosis disease survivors with abnormal spirometry,[7,53] symptoms might be due to non-pulmonary conditions (eg, deconditioning, or cardiac disease), individuals might under-report symptoms, and symptom reporting is subjective and can vary over time and between individuals.

There has been wide heterogeneity in the types of symptoms evaluated in post-tuberculosis lung disease studies, including the presence or absence of cough, dyspnoea or breathlessness, wheezing, sputum production, fatigue, haemoptysis, and chest pain.[6,13] Various instruments have been used to evaluate symptoms and health status, including the modified Medical Research Council questionnaire, St George's Research Questionnaire, 36-item Short Form Health Survey (a quality-of-life measure), and the COPD Assessment Test.[13,54] To our knowledge, data validating the relationship between these symptoms or symptom scores and clinically meaningful outcomes are scarce. One study found that the presence of respiratory symptoms at tuberculosis disease treatment completion was the strongest predictor of worsening spirometry, health seeking, and symptoms or functional limitations at 1-year post-treatment,[7] but further work is needed to

validate the relationship between symptoms and outcomes in post-tuberculosis lung disease.

Given this heterogeneity and absence of validation, it was felt that there were inadequate data to designate a particular instrument for respiratory symptom assessment in post-tuberculosis lung disease. However, the case definition lists a finite number of respiratory symptoms that are commonly associated with pulmonary disease: cough, dyspnoea or breathlessness, wheezing, and limitation in activities attributed to respiratory impairment. Emphasis has been placed on the self-reporting of functional limitations as a patient-centred and meaningful outcome that could identify individuals who do not report resting symptoms.

To increase the likelihood that respiratory symptoms are indeed sequelae of tuberculosis disease, any symptoms satisfying the case definition must have either began or worsened during the period of active tuberculosis disease or thereafter (ie, they are temporally attributable to tuberculosis disease). Symptoms that begin long after tuberculosis disease treatment completion could be due to residual tuberculosis pathology or to non-tuberculosis disease causes and likely will require careful clinical evaluation to adjudicate a possible causal relationship to tuberculosis disease.

Because post-tuberculosis lung disease is considered a chronic condition, the case definition requires the presence of respiratory symptoms for at least 8 weeks in adults and 4 weeks in children at the time of assessment for post-tuberculosis lung disease. This duration is consistent with common definitions of chronic respiratory complaints, including cough.[55,56] It also allows for post-tuberculosis lung disease to be defined at the end of tuberculosis disease treatment; a timepoint that enables evaluation by tuberculosis programmes in the final weeks of antitubercular therapy. The timing of symptom evaluations should be reported in the reporting framework.

Although exercise testing, general symptom, and overall quality of life questionnaires are valuable for clinical evaluations, these measures were not included in the case definition due to their non-specificity for respiratory disease and a desire to maximise simplicity in the absence of robust validation.

Imaging criteria

See Online for appendix

It is anticipated that most imaging assessments will be done by chest x-ray, given the greater cost and limited availability of more advanced technologies, such as CT imaging, in LMICs. Although CT provides greater structural detail than chest x-ray, the advantages in providing diagnostic or prognostic information in post-tuberculosis lung disease remain unclear. We therefore accept the use of either CT or chest x-ray to identify imaging abnormalities in this case definition. There is a paucity of data on the accuracy of chest ultrasound for identifying post-tuberculosis disease changes and this modality is therefore not included in the definition.

A variety of radiographical abnormalities can be seen on chest x-ray and CT after tuberculosis disease treatment, including pleural, parenchymal, and airway pathologies.[8] Specific chest x-ray findings at tuberculosis disease diagnosis have been associated with microbiological and tuberculosis disease treatment outcomes.[57] However, there is a paucity of data relating radiographical findings at tuberculosis disease diagnosis to lung function and residual post-tuberculosis lung disease.[58-60] Diverse approaches to reporting post-tuberculosis disease imaging have been used in the literature, with little validation against patient respiratory health outcomes,[8] and approaches to image interpretation are rapidly evolving with the use of artificial intelligence software.[61-63] Given the broad range of imaging findings and reporting approaches in the literature and the scarcity of data on their association with long-term outcomes, the case definition allows individual clinical investigators to decide which radiological abnormalities of the parenchyma, airways, or pleura to use to define disease. Conceptually, these should be changes that are consistent with previous tuberculosis disease and could explain other aspects of post-tuberculosis lung disease if present (ie, symptoms and lung function). Details of the patterns chosen or criteria used to establish whether imaging abnormalities consistent with post-tuberculosis lung disease are present should be included in the reporting framework.

(3) Radiographical, functional, and symptomatic pulmonary disease manifestations should be attributable, at least in part, to previous tuberculosis disease

The radiographical, functional, and symptomatic pulmonary disease manifestations should be attributable at least in part to previous tuberculosis disease. Although health before tuberculosis disease is often unknown to investigators, studies can include questions about previous respiratory disease and these should be reported in the reporting framework. Evaluation of alternative explanations for respiratory signs and symptoms and attribution of disease to tuberculosis can be challenging and ideally would be done by a clinician investigator. A clinical approach to alternative explanations is similar to that used for case definitions of tuberculosis-associated immune reconstitution inflammatory syndrome and tuberculosis meningitis.[18,19] Case examples using the proposed case definition are presented in the appendix (pp 5, 6).

Certainty grading

Although classifying individuals as having possible, probable, or confirmed post-tuberculosis lung disease was rated as desirable by attendees at the 2nd Symposium, it was ultimately felt that the inclusion of a certainty determination would introduce additional complexity into the definition without improving specificity, thus limiting its ease of use and uptake.

Severity grading

Three parameters are proposed for severity grading of post-tuberculosis lung disease. First is a lung function severity grading; grading metrics, including those for spirometry (eg, FEV_1, FVC, and FEV_1 to FVC ratio), diffusing capacity, and volumes (eg, total lung capacity, functional residual capacity, and residual volume), should be reported using the ERS–ATS three-level system, with Z-scores between –1·65 and –2·50 as mild, between –2·51 and –4·00 as moderate, and less than or equal to –4·10 as severe.[29]

Second is symptom severity grading. The mMRC score is recommended for grading the severity of symptomatic impairment because breathlessness is relevant to wellbeing and its grading and assessment are simple. The mMRC score reflects the person's experience of breathlessness, which is associated with both mortality and quality of life for a range of cardiopulmonary diseases.[40,64] It is a widely used metric, including in the Global Initiative for Chronic Obstructive Lung Disease guidelines for COPD grading, and is both simple and feasible to implement in LMICs.[24,65] Scores should be reported as 0 (breathless only on strenuous exercise), 1 (short of breath when hurrying on the level or walking up a slight hill), 2 (walking slower than people of the same age on the level because of breathlessness), 3 (stopping for breath after walking about 100 m or a few minutes on the level), or 4 (too breathless to leave the house or breathless when dressing or undressing).

Third is hypoxaemia. Chronic resting hypoxaemia is a sign of advanced lung disease, predicts mortality, is more common among individuals with previous tuberculosis disease compared with those with no history of tuberculosis disease, and is treatable with supplemental oxygen, with benefits to survival and quality of life.[66–68] Exertional hypoxaemia is also common among individuals with very low lung function and is associated with worse quality of life and increased mortality (table 2).[69–71] Thus, we recommend assessing for and reporting the proportion of individuals with resting hypoxaemia, which we define as a peripheral oxygen saturation measured by pulse oximetry of less than 90% or a partial pressure of oxygen in arterial blood of less than 60 mm Hg on room air at one timepoint, given that a single assessment is simple and pragmatic. When feasible, chronic resting hypoxaemia—established with at least two measurements taken at least 4 weeks apart—and exertional hypoxemia—defined by a desaturation to less than 90% when ambulating or during exercise testing—should also be reported.

Reporting framework

Panel 2 shows the proposed reporting framework for post-tuberculosis lung disease studies, which can be included in the primary manuscript or in supplemental files. The goal of the reporting framework is to encourage authors to be explicit about how a post-tuberculosis lung disease assessment has been carried out, using a simple template to support evidence synthesis.

Discussion and future considerations

Key components of this post-tuberculosis lung disease research case definition include an assessment of the certainty of previous pulmonary or pleural tuberculosis disease, a focus on individuals with current evidence of pulmonary disease, and the absence of an alternative explanation for this disease. The definition aims to identify those with disease that is likely to be associated with long-term morbidity and mortality by requiring abnormalities in at least two clinical domains. It is accompanied by a comprehensive reporting framework (panel 2), which is intended to improve the harmonisation of results across different studies and settings. In summary, the definition proposes that post-tuberculosis lung disease is defined as evidence of pulmonary disease in at least two of three domains (lung function, respiratory symptoms, and imaging) that is attributable at least in part to previous tuberculosis disease in an individual who had previous bacteriologically confirmed or clinically diagnosed pulmonary or pleural tuberculosis disease and who does not currently have evidence of tuberculosis disease.

We acknowledge that the case definition and reporting framework proposed here are a starting point for post-tuberculosis lung disease research. We expect that they will evolve over time and will require updating as more data and evidence emerge, particularly for newer assessments of lung function, symptoms, or imaging. Longitudinal data on the relationship between the parameters included in this definition (lung function, respiratory symptoms, and imaging findings) measured at or after tuberculosis disease treatment completion and outcomes, including morbidity, mortality, and health-care use, will be crucial to this evolution. There is a particular need for chest x-ray and CT scoring systems for post-tuberculosis lung disease that are validated against clinical outcomes.

Strengths of this definition include that it has been developed with broad input from clinicians, people with a history of tuberculosis disease, researchers, and public health specialists. It is not intended to be prescriptive and can be used alongside researcher-driven disease measures focused on specific research questions. The accompanying reporting framework will allow for transparency and evidence synthesis. Limitations are described in the sections above and importantly include the absence of data validating this definition with respect to disease course and outcomes.

In conclusion, greater consistency and harmonisation of post-tuberculosis lung disease studies with this case definition and reporting framework has the potential to provide a stronger evidence base from which to improve care for millions of tuberculosis survivors globally.

Search strategy and selection criteria

References for this Personal View were identified by the authors and focused on systematic reviews of post-tuberculosis lung disease, clinical guidelines for post-tuberculosis care, evidence presented at the 1st and 2nd International Post-Tuberculosis Symposia, and their experiences in the field. There were no restrictions for date or language of publication for the cited literature. We did not conduct a formal literature search specifically for this Personal View because a 2023 systematic review described variability in post-tuberculosis lung disease definitions used in research.[6]

Contributors

GPB, BA, AB, GG, CK, NN, RN, MMvdZ, JM, and SA developed the concept and contributed to writing and review of the manuscript. All authors contributed to the literature search. GPB wrote the first draft of the manuscript. IS provided input on tuberculosis survivors' perspectives. All authors approved the final version.

Declaration of interests

The authors declare no competing interests.

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